# DriftGuard Clinical | Movement Risk Screening Report

AI-Assisted Biomechanical Assessment & Baseline Comparison Platform

***NON-DIAGNOSTIC SCREENING AID:*** *This report presents computer-vision movement screening proxies and baseline comparisons. It is intended solely for clinical decision support by licensed doctors and physiotherapists, and does NOT constitute a formal medical diagnosis or ACL injury confirmation.*

***MEASUREMENT PROVENANCE & HARDWARE DISCLOSURE:*** *Camera Setup: Monocular RGB (MediaPipe 3D) | IMU Status: IMU integration available — hardware not connected | Kinematics: Pose-Derived Proxies | Mocap Validation: Not Validated against Mocap Reference*

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## 1. Athlete & Session Metadata

**Athlete Name:** Demo Athlete

**Session ID:** ses_698298c0c869

**Athlete ID:** ath_6f7a056c760f

**Assessment Date:** 2026-10-06

**Sport Profile:** Basketball

**Camera Setup:** frontal+sagittal

**Dominant Leg:** Unknown

**Overall Risk Band:** **LOW**

## 2. Observed Kinematic Summary (Mean / Event Peak)

| Biomechanical Metric | Left Limb | Right Limb | Limb Symmetry (LSI) |
|---|---|---|---|
| Knee FPPA Valgus Proxy | -19.8° | 7.8° | 200.0% |
| Sagittal Knee Flexion | 19.8° | 7.8° | 86.8% |
| Trunk Lateral Lean | 52.0° | N/A | N/A |

## 3. Confirmed Risk Flags & Literature Evidence

| Feature | Risk Band | Deviation | Clinical Detail / Citation |
|---|---|---|---|
| knee_flexion_left | **Moderate** | +-1.8 IQR | **Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 41.0 degrees (+/-1.8 degrees) by 1.8x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor) |

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| knee_flexion_left | **Moderate** | +-1.8 IQR | **Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 41.0 degrees (+/-1.8 degrees) by 1.8x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor) |
| fppa_right | **High** | +-3.5 IQR | **Elevated movement-risk pattern: Right knee valgus angle (fppa) has been substantially elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.5 degrees (+/-1.6 degrees) by 3.5x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor) |
| knee_flexion_left | **Moderate** | +-7.7 IQR | **Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 40.4 degrees (+/-1.8 degrees) by 7.7x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor) |
| fppa_right | **High** | +-3.6 IQR | **Elevated movement-risk pattern: Right knee valgus angle (fppa) has been substantially elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.7 degrees (+/-1.6 degrees) by 3.6x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor) |
| knee_flexion_left | **Moderate** | +-1.8 IQR | **Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 40.4 degrees (+/-7.6 degrees) by 1.8x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor) |

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|---|---|---|---|
| fppa_right | **High** | +-2.9 IQR | **Elevated movement-risk pattern: Right knee valgus angle (fppa) has been substantially elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.4 degrees (+/-1.9 degrees) by 2.9x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor) |
| knee_flexion_left | **Moderate** | +-1.0 IQR | **Elevated movement-risk pattern: Left knee landing flexion has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in left knee landing flexion, consistently below their established baseline median of 40.4 degrees (+/-13.3 degrees) by 1.0x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Shin et al. 2009 (reduced landing knee flexion as ACL-injury risk factor) |
| fppa_right | **Moderate** | +-2.9 IQR | **Elevated movement-risk pattern: Right knee valgus angle (fppa) has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 7.4 degrees (+/-1.9 degrees) by 2.9x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor) |
| fppa_right | **Moderate** | +-2.5 IQR | **Elevated movement-risk pattern: Right knee valgus angle (fppa) has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 5.7 degrees (+/-1.9 degrees) by 2.5x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor) |
| fppa_right | **Moderate** | +-2.6 IQR | **Elevated movement-risk pattern: Right knee valgus angle (fppa) has been moderately elevated across recent sessions.** This athlete's recent movement trials show a biomechanical deviation in right knee valgus angle (FPPA), consistently below their established baseline median of 6.0 degrees (+/-1.9 degrees) by 2.6x their normal spread. This pattern has persisted across at least 3 of their last 5 recorded events. Citation: Hewett et al. 2005; Bates et al. 2020 (knee abduction / valgus angle as ACL-injury risk factor) |

## 4. Clinician Observations & Neuromuscular Retraining Notes

*No clinician notes recorded for this session yet.*

**Assessing Clinician Signature: _____** **Date: _____**